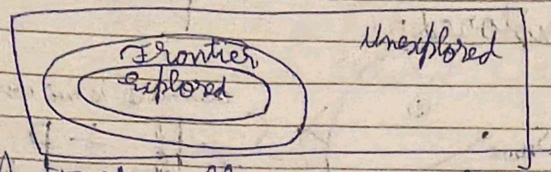
24/08/26

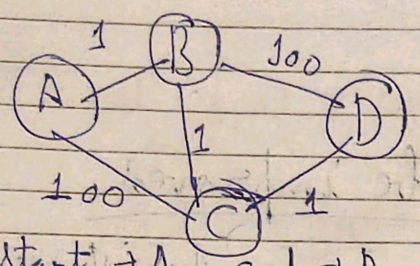
like dijikshtra's

UCS (Uniform Cost Search)



Initially all are unexplored.

Eg:



Start → A, end → D

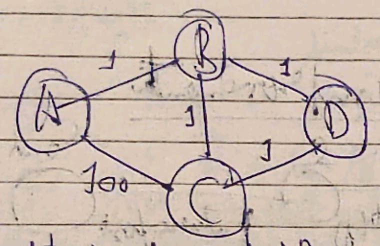
min ⇒ A → B → C → D
cost = 3

If touched at least once

U	F	E
→ A	~~A, 0~~	A
B	~~B, 1~~	B, 1
C	~~C, 100~~ 2	C, 2
D	~~D, 101~~ 3	D, 3

If touched at least once, state moves to frontier.

Eg:



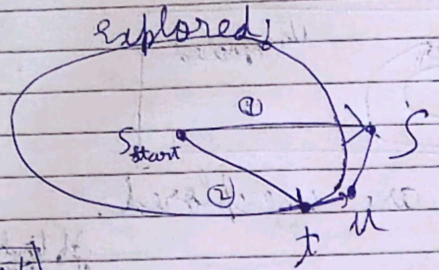
Start → A, end → D

min ⇒ A → B → D
cost = 2

U	F popped	E
→ A	~~A, 0~~	A, 0
B	~~B, 1~~	B, 1
C	~~C, 100~~ 2	D, 2
D	~~D, 2~~	

drawback of UCS: doesn't work for -ve edge costs

→ Analysis of UCS



P_s

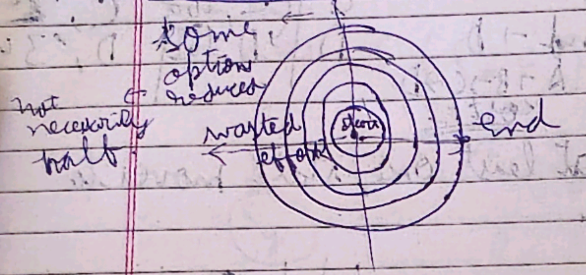
assume ② is min cost.

$P_{s_{start}} + P_t ~~+ P(t,u)~~ + P_{tu} \geq P_u \geq P_s$

claimed a/c to assumptio

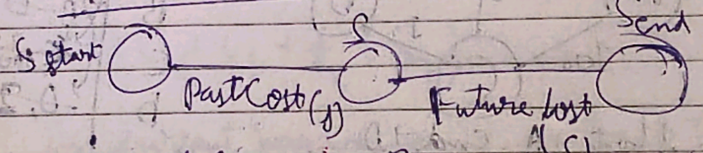
UCS → priority queue
↓
O(n log n)

→ Can UCS be improved



heuristic search
h(n)

Possible Improvement



PastCost(s) → PastCost(s) + FutureCost(s)
↓
PastCost(s) + h(s)

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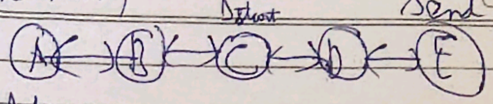
{same edge costs}

Problem:



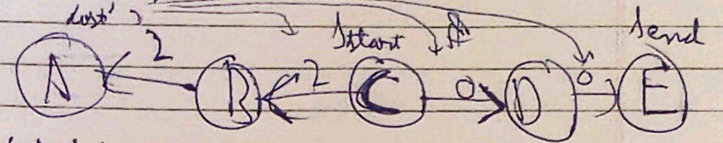
If C → B is taken it is unnecessary effort we need to prevent C → B.

· A* search

→ Run UCS with modified edge costs

All initial are s cost.

$$\text{Cost}'(s,a) = \text{cost}(s,a) + h(\overset{\text{successor state}}{\text{succ}(s,a)}) - h($$



← $h(s) = 4$ 3 2 1 0

can be defined on own

{have started from goal and checked cost to reach each node from goal

Assume successor → D

$\text{Cost}'([illegible]) = 1 + h(D) - h(C)$
$= 1 + 1 - 2 = 0$

Similarly can assume B also.
D gives minⁿ.

Cost' is only to guide C to D. Actual cost is still not changed.
Minᵐ cost is still 2.